Elements of Information Theory **2026-2027**

Lecture 1: Information Measures

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Outline of the course:

- Information measures
- Method of types
- Lossless source coding
- Lossy source coding
- Channel coding
- Information theory & statistics / machine learning

Notations:

- log: base 2; $\ln x = \log x / \log e$.
- $\mathcal{X}, \mathcal{Y}, \mathcal{Z}$: alphabets.
- $\mathcal{P}(\mathcal{X})$: set of probability distributions over $\mathcal{X}$.
- $\mathrm{P}, \mathrm{Q} \in \mathcal{P}(\mathcal{X})$: probability measures; p, q: their densities (pdf: Lebesgue measure; pmf: counting measure).
- $\mathrm{P} \ll \mathrm{Q}$: P is **absolutely continuous** with respect to Q, i.e.,

in general:	$\mathrm{Q}(\mathcal{A}) = 0 \Rightarrow \mathrm{P}(\mathcal{A}) = 0$ for every event $\mathcal{A}$,
pmfs:	$q(x) = 0 \Rightarrow p(x) = 0$ for every x,
pdfs:	$q(x) = 0 \Rightarrow p(x) = 0$ for almost every x.

  Then p/q (in general, the Radon–Nikodym derivative dP/dQ) is well defined P-almost surely.
- $\mathrm{X}, \mathrm{Y}, \mathrm{Z}, \mathrm{U}, \mathrm{V}, \mathrm{S}$: random variables.
- $\mathrm{P_X}$, $\mathrm{Q_X}$: distributions of X (see below); $p(x) \equiv p_{\mathrm{X}}(x)$.
- $\mathrm{P_{Y|X}}$: transition probability kernel (channel), i.e., a family of distributions in $\mathcal{P}(\mathcal{Y})$ indexed by $x \in \mathcal{X}$,

$$\mathrm{P_{Y|X}} := \big(\underbrace{\mathrm{P_{Y|X}}=x}_{\text{probability measure of Y}}\big)_{x\in\mathcal{X}}, \qquad p(y\,|\,x) \equiv \mathrm{P_{Y|X}}=x(y) \equiv \mathrm{P}(y\,|\,\mathrm{X}=x).$$

- $\mathcal{P}(\mathcal{Y}|\mathcal{X})$: set of channels from $\mathcal{X}$ to $\mathcal{Y}$; $\quad \mathrm{P}_{\mathrm{Y}|\mathrm{X}} \in \mathcal{P}(\mathcal{Y}|\mathcal{X})$.
- For any $\mathrm{P}_{\mathrm{X}} \in \mathcal{P}(\mathcal{X})$ and $\mathrm{P}_{\mathrm{Y}|\mathrm{X}} \in \mathcal{P}(\mathcal{Y}|\mathcal{X})$, the joint distribution of (X, Y) and the output distribution of the channel are

$$\begin{aligned} \mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}} \equiv \mathrm{P}_{\mathrm{Y}|\mathrm{X}}\mathrm{P}_{\mathrm{X}} &\in \mathcal{P}(\mathcal{X} \times \mathcal{Y}), \qquad \text{with density } p(x)\,p(y\,|\,x), \\ \mathrm{P}_{\mathrm{Y}|\mathrm{X}} \circ \mathrm{P}_{\mathrm{X}} := \mathbb{E}_{\tilde{\mathrm{X}} \sim \mathrm{P}_{\mathrm{X}}}\big[\mathrm{P}_{\mathrm{Y}|\mathrm{X}=\tilde{\mathrm{X}}}\big] &\in \mathcal{P}(\mathcal{Y}), \qquad \text{the Y-marginal of } \mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}}. \end{aligned}$$

  The output distribution is a **mixture** of the distributions $\mathrm{P}_{\mathrm{Y}|\mathrm{X}=x}$, $x \in \mathcal{X}$, with mixing distribution P_{X}.

One random variable, several distributions. A random variable X with values in $\mathcal{X}$ is often considered under several distributions $\mathrm{P}_{\mathrm{X}}, \mathrm{Q}_{\mathrm{X}} \in \mathcal{P}(\mathcal{X})$, for instance the true distribution of a source and a model of it. Formally, X can be taken as the identity map on $\mathcal{X}$, so that choosing a distribution in $\mathcal{P}(\mathcal{X})$ is choosing the distribution of X. Each distribution is therefore defined explicitly, in order (P_{X}, then $\mathrm{P}_{\mathrm{Y}|\mathrm{X}}$, then $\mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}}$ and $\mathrm{P}_{\mathrm{Y}|\mathrm{X}} \circ \mathrm{P}_{\mathrm{X}}$), and the notation says which one is used, e.g., $\mathrm{X} \sim \mathrm{Q}_{\mathrm{X}}$ or $\mathbb{E}_{\mathrm{P}}[\,\cdot\,]$.

Example (binary symmetric channel). Let $\mathcal{X} = \mathcal{Y} = \{0, 1\}$ and $p, q, \varepsilon \in [0, 1]$. We define, in this order,

- the input distribution $\mathrm{P}_{\mathrm{X}} = \mathrm{Bern}(p)$, i.e., $p(1) = p$ and $p(0) = 1 - p$;
- the channel $\mathrm{P}_{\mathrm{Y}|\mathrm{X}} = \big(\mathrm{Bern}(\varepsilon), \mathrm{Bern}(1-\varepsilon)\big)$, i.e., $\mathrm{P}_{\mathrm{Y}|\mathrm{X}=0} = \mathrm{Bern}(\varepsilon)$ and $\mathrm{P}_{\mathrm{Y}|\mathrm{X}=1} = \mathrm{Bern}(1-\varepsilon)$: the input bit is flipped with probability ε;
- the joint distribution $\mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}}$, whose pmf $p(x)\,p(y\,|\,x)$ is given in the table below;
- the output distribution $\mathrm{P}_{\mathrm{Y}} := \mathrm{P}_{\mathrm{Y}|\mathrm{X}} \circ \mathrm{P}_{\mathrm{X}} = (1-p)\mathrm{Bern}(\varepsilon) + p\mathrm{Bern}(1-\varepsilon) = \mathrm{Bern}(p \star \varepsilon)$, with $p \star \varepsilon := p(1-\varepsilon) + (1-p)\varepsilon$, given by the column sums of the table;
- the reverse channel $\mathrm{P}_{\mathrm{X}|\mathrm{Y}}$, with pmf $p(x\,|\,y) = p(x)\,p(y\,|\,x)/p(y)$, so that $\mathrm{P}_{\mathrm{Y}}\mathrm{P}_{\mathrm{X}|\mathrm{Y}} = \mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}}$; for example, $\mathrm{P}_{\mathrm{X}|\mathrm{Y}=1} = \mathrm{Bern}\big(p(1-\varepsilon)/(p \star \varepsilon)\big)$.

	$y = 0$	$y = 1$	P_{X}
$x = 0$	$(1-p)(1-\varepsilon)$	$(1-p)\varepsilon$	$1-p$
$x = 1$	$p\varepsilon$	$p(1-\varepsilon)$	p
P_{Y}	$1 - p \star \varepsilon$	$p \star \varepsilon$	

With another input distribution $\mathrm{Q}_{\mathrm{X}} = \mathrm{Bern}(q)$, the same channel gives another joint distribution $\mathrm{Q}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}}$ and another output distribution $\mathrm{P}_{\mathrm{Y}|\mathrm{X}} \circ \mathrm{Q}_{\mathrm{X}} = \mathrm{Bern}(q \star \varepsilon)$. For $p = \varepsilon = \frac{1}{2}$, every entry of the table is $\frac{1}{4}$: X and Y are independent and uniform.

1.1 Convex functions and Jensen's inequality

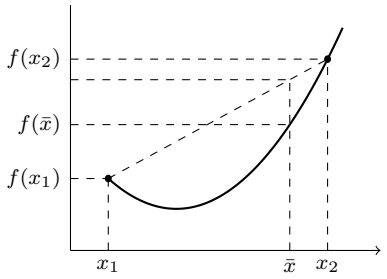
A set $\mathcal{X}$ (in a vector space) is a **convex set** if

$$\forall\, x_1, x_2 \in \mathcal{X},\ \lambda \in [0, 1]: \qquad \lambda x_1 + (1-\lambda) x_2 \in \mathcal{X}.$$

For example, $\mathcal{P}(\mathcal{X})$ is a convex set.

A function $f\colon \mathcal{X} \to \mathbb{R}$ on a convex set $\mathcal{X}$ is a **convex function** if

$$\forall\, x_1, x_2 \in \mathcal{X},\ \lambda \in [0,1]: \qquad \underbrace{f\big(\lambda x_1 + (1-\lambda)x_2\big)}_{f(\bar{x})} \le \underbrace{\lambda f(x_1) + (1-\lambda) f(x_2)}_{\bar{f}}.$$



f is a **concave** function if $-f$ is convex.

Jensen's inequality. (Jensen)

For any convex function f and any distribution P of X,

$$\mathbb{E}_{\mathrm{P}}\big[f(\mathrm{X})\big] \ge f\big(\mathbb{E}_{\mathrm{P}}[\mathrm{X}]\big).$$

If f is concave, the inequality is reversed:

$$\mathbb{E}_{\mathrm{P}}\big[f(\mathrm{X})\big] \le f\big(\mathbb{E}_{\mathrm{P}}[\mathrm{X}]\big).$$

Examples (check the sign of f''): $x \mapsto \log x$ is concave on $x > 0$, and $x \mapsto x \log x$ is convex on $x > 0$.

1.2 Information measures

- Entropy
- Cross-entropy
- Divergence
- Mutual information

1.2.1 Entropy

Let $\mathrm{X} \in \mathcal{X}$ be discrete, $\mathrm{X} \sim \mathrm{P}_{\mathrm{X}}$.

- $H(\mathrm{X}) \equiv H(\mathrm{P}_{\mathrm{X}}) := \mathbb{E}_{\mathrm{X}\sim\mathrm{P}_{\mathrm{X}}}\Big[\log \dfrac{1}{p(\mathrm{X})}\Big] = \displaystyle\sum_{x\in\mathcal{X}} p(x) \log \frac{1}{p(x)}$.

For $(\mathrm{X}, \mathrm{Y}) \in \mathcal{X} \times \mathcal{Y}$ with $(\mathrm{X}, \mathrm{Y}) \sim \mathrm{P}_{\mathrm{XY}}$, the joint and conditional entropies are

- $H(\mathrm{X},\mathrm{Y}) \equiv H(\mathrm{P}_{\mathrm{XY}}) := \mathbb{E}_{\mathrm{P}_{\mathrm{XY}}}\Big[\log \dfrac{1}{p(\mathrm{X},\mathrm{Y})}\Big]$,
- $H(\mathrm{X}\,|\,\mathrm{Y}) \equiv H(\mathrm{P}_{\mathrm{X|Y}} \mid \mathrm{P}_{\mathrm{Y}}) := \mathbb{E}_{\bar{\mathrm{Y}}\sim\mathrm{P}_{\mathrm{Y}}}\big[H(\mathrm{P}_{\mathrm{X|Y}=\bar{\mathrm{Y}}})\big] = \mathbb{E}_{\mathrm{P}_{\mathrm{XY}}}\Big[\log \dfrac{1}{p(\mathrm{X}\,|\,\mathrm{Y})}\Big]$.

"Chain rule":

$$H(\mathrm{P}_{\mathrm{X|Y}}\mathrm{P}_{\mathrm{Y}}) = H(\mathrm{P}_{\mathrm{Y}}) + H(\mathrm{P}_{\mathrm{X|Y}} \mid \mathrm{P}_{\mathrm{Y}}),$$

i.e., in the conventional notation, $H(\mathrm{X},\mathrm{Y}) = H(\mathrm{Y}) + H(\mathrm{X}\,|\,\mathrm{Y})$. By induction, for X^n,

$$\begin{aligned} & H(\mathrm{X}^n) = H(\mathrm{X}^{n-1}) + H(\mathrm{X}_n\,|\,\mathrm{X}^{n-1}) \\ \Longrightarrow \quad & H(\mathrm{X}^n) = \sum_{i=1}^{n} H(\mathrm{X}_i\,|\,\mathrm{X}^{i-1}), \qquad \text{with } \mathrm{X}^n \equiv [\mathrm{X}_1,\ldots,\mathrm{X}_n]. \end{aligned}$$

1.2.2 Cross-entropy

For $\mathcal{X}$ discrete, $\mathrm{P},\mathrm{Q} \in \mathcal{P}(\mathcal{X})$ and $\mathrm{X} \sim \mathrm{P}$,

$$H(\mathrm{P}\|\mathrm{Q}) := \mathbb{E}_{\mathrm{P}}\Big[\log \frac{1}{q(\mathrm{X})}\Big] = \sum_{x\in\mathcal{X}} p(x) \log \frac{1}{q(x)} \qquad (+\infty \text{ if } \mathrm{P} \not\ll \mathrm{Q}),$$

and, for conditional distributions,

$$H(\mathrm{P}_{\mathrm{X|Y}}\|\mathrm{Q}_{\mathrm{X|Y}} \mid \mathrm{P}_{\mathrm{Y}}) := \mathbb{E}_{\bar{\mathrm{Y}}\sim\mathrm{P}_{\mathrm{Y}}}\big[H(\mathrm{P}_{\mathrm{X|Y}=\bar{\mathrm{Y}}}\|\mathrm{Q}_{\mathrm{X|Y}=\bar{\mathrm{Y}}})\big] = \mathbb{E}_{\mathrm{P}_{\mathrm{X|Y}}\mathrm{P}_{\mathrm{Y}}}\Big[\log \frac{1}{q(\mathrm{X}\,|\,\mathrm{Y})}\Big].$$

"Chain rule":

$$\begin{aligned} H(\mathrm{P}_{\mathrm{XY}}\|\mathrm{Q}_{\mathrm{XY}}) &= H(\mathrm{P}_{\mathrm{X|Y}}\mathrm{P}_{\mathrm{Y}}\|\mathrm{Q}_{\mathrm{X|Y}}\mathrm{Q}_{\mathrm{Y}}) = H(\mathrm{P}_{\mathrm{Y}}\|\mathrm{Q}_{\mathrm{Y}}) + H(\mathrm{P}_{\mathrm{X|Y}}\|\mathrm{Q}_{\mathrm{X|Y}} \mid \mathrm{P}_{\mathrm{Y}}), \\ H(\mathrm{P}_{\mathrm{X}^n}\|\mathrm{Q}_{\mathrm{X}^n}) &= H(\mathrm{P}_{\mathrm{X}_n\mathrm{X}^{n-1}}\|\mathrm{Q}_{\mathrm{X}_n\mathrm{X}^{n-1}}) \\ &= H(\mathrm{P}_{\mathrm{X}^{n-1}}\|\mathrm{Q}_{\mathrm{X}^{n-1}}) + H(\mathrm{P}_{\mathrm{X}_n|\mathrm{X}^{n-1}}\|\mathrm{Q}_{\mathrm{X}_n|\mathrm{X}^{n-1}} \mid \mathrm{P}_{\mathrm{X}^{n-1}}) \\ &= \sum_{i=1}^{n} H(\mathrm{P}_{\mathrm{X}_i|\mathrm{X}^{i-1}}\|\mathrm{Q}_{\mathrm{X}_i|\mathrm{X}^{i-1}} \mid \mathrm{P}_{\mathrm{X}^{i-1}}). \end{aligned}$$

1.2.3 Differential entropy and cross-entropy

For X continuous with density p,

$$h(\mathrm{X}) \equiv h(\mathrm{P}_{\mathrm{X}}) := \mathbb{E}_{\mathrm{P}}\Big[\log \frac{1}{p(\mathrm{X})}\Big].$$

Examples: uniform and Gaussian distributions. Unlike H, h can be negative ($h(\mathrm{X}) \not\geq 0$ in general) and is "not scale-free": $h(a\mathrm{X}) = h(\mathrm{X}) + \log|a|$. The differential cross-entropy $h(\mathrm{P}\|\mathrm{Q})$ and the conditional versions, such as $h(\mathrm{P}_{\mathrm{X|Y}}\|\mathrm{Q}_{\mathrm{X|Y}} \mid \mathrm{P}_{\mathrm{Y}})$, are defined in the same way with densities, and the chain rules hold.

1.2.4 Kullback–Leibler divergence

The **Kullback–Leibler (KL) divergence**, or **relative entropy**, is

$$D(\mathrm{P}\|\mathrm{Q}) := \mathbb{E}_{\mathrm{P}}\Big[\log\frac{p(\mathrm{X})}{q(\mathrm{X})}\Big] \qquad (+\infty \text{ if } \mathrm{P} \not\ll \mathrm{Q}).$$

If $\mathcal{X}$ is discrete, $D(\mathrm{P}\|\mathrm{Q}) = H(\mathrm{P}\|\mathrm{Q}) - H(\mathrm{P})$ whenever $H(\mathrm{P}) < \infty$; if P and Q have pdfs, $D(\mathrm{P}\|\mathrm{Q}) = h(\mathrm{P}\|\mathrm{Q}) - h(\mathrm{P})$ whenever $h(\mathrm{P})$ is finite.

For joint distributions,

$$\begin{aligned} D(\mathrm{P}_{\mathrm{XY}}\|\mathrm{Q}_{\mathrm{XY}}) &= \mathbb{E}_{\mathrm{P}_{\mathrm{XY}}}\Big[\log\frac{p(\mathrm{X})}{q(\mathrm{X})}\Big] + \mathbb{E}_{\mathrm{P}_{\mathrm{XY}}}\Big[\log\frac{p(\mathrm{Y}\,|\,\mathrm{X})}{q(\mathrm{Y}\,|\,\mathrm{X})}\Big] \\ &= D(\mathrm{P}_{\mathrm{X}}\|\mathrm{Q}_{\mathrm{X}}) + \mathbb{E}_{\bar{\mathrm{X}}\sim\mathrm{P}_{\mathrm{X}}}\big[D(\mathrm{P}_{\mathrm{Y}|\mathrm{X}=\bar{\mathrm{X}}}\|\mathrm{Q}_{\mathrm{Y}|\mathrm{X}=\bar{\mathrm{X}}})\big], \end{aligned}$$

where the conditional divergence is

$$D(\mathrm{P}_{\mathrm{Y}|\mathrm{X}}\|\mathrm{Q}_{\mathrm{Y}|\mathrm{X}} \mid \mathrm{P}_{\mathrm{X}}) := \mathbb{E}_{\bar{\mathrm{X}}\sim\mathrm{P}_{\mathrm{X}}}\big[D(\mathrm{P}_{\mathrm{Y}|\mathrm{X}=\bar{\mathrm{X}}}\|\mathrm{Q}_{\mathrm{Y}|\mathrm{X}=\bar{\mathrm{X}}})\big].$$

Then, we get the "chain rule"

$$D(\mathrm{P}_{\mathrm{Y}|\mathrm{X}}\mathrm{P}_{\mathrm{X}}\|\mathrm{Q}_{\mathrm{Y}|\mathrm{X}}\mathrm{Q}_{\mathrm{X}}) = D(\mathrm{P}_{\mathrm{X}}\|\mathrm{Q}_{\mathrm{X}}) + D(\mathrm{P}_{\mathrm{Y}|\mathrm{X}}\|\mathrm{Q}_{\mathrm{Y}|\mathrm{X}} \mid \mathrm{P}_{\mathrm{X}})$$

and, by induction,

$$D(\mathrm{P}_{\mathrm{X}^n}\|\mathrm{Q}_{\mathrm{X}^n}) = \sum_{i=1}^{n} D(\mathrm{P}_{\mathrm{X}_i|\mathrm{X}^{i-1}}\|\mathrm{Q}_{\mathrm{X}_i|\mathrm{X}^{i-1}} \mid \mathrm{P}_{\mathrm{X}^{i-1}}).$$

1.2.5 Mutual information

$$\begin{aligned} I(\mathrm{X};\mathrm{Y}) \equiv I(\mathrm{P}_{\mathrm{XY}}) &\equiv I(\mathrm{P}_{\mathrm{X}}, \mathrm{P}_{\mathrm{Y}|\mathrm{X}}) \equiv I(\mathrm{P}_{\mathrm{Y}}, \mathrm{P}_{\mathrm{X}|\mathrm{Y}}) \\ &:= D(\mathrm{P}_{\mathrm{XY}}\|\underbrace{\mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}}}_{\text{ind. distributions}}) \\ &= D(\mathrm{P}_{\mathrm{X}|\mathrm{Y}}\|\mathrm{P}_{\mathrm{X}} \mid \mathrm{P}_{\mathrm{Y}}) \\ &= D(\mathrm{P}_{\mathrm{Y}|\mathrm{X}}\|\mathrm{P}_{\mathrm{Y}} \mid \mathrm{P}_{\mathrm{X}}), \end{aligned}$$

where $\mathrm{P}_{\mathrm{XY}} = \mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}} = \mathrm{P}_{\mathrm{Y}}\mathrm{P}_{\mathrm{X}|\mathrm{Y}}$, and P_{X}, P_{Y} are the marginals of P_{XY}. Similarly, the conditional mutual information is

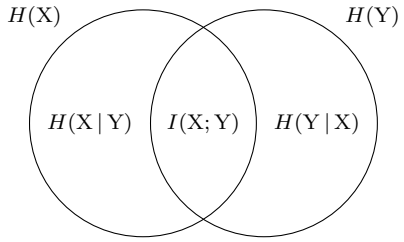
$$\begin{aligned} I(\mathrm{X};\mathrm{Y}\,|\,\mathrm{Z}) &\equiv I(\mathrm{P}_{\mathrm{XY}|\mathrm{Z}} \mid \mathrm{P}_{\mathrm{Z}}) \\ &:= D(\mathrm{P}_{\mathrm{XY}|\mathrm{Z}}\|\mathrm{P}_{\mathrm{X}|\mathrm{Z}}\mathrm{P}_{\mathrm{Y}|\mathrm{Z}} \mid \mathrm{P}_{\mathrm{Z}}) \\ &= D(\mathrm{P}_{\mathrm{X}|\mathrm{YZ}}\|\mathrm{P}_{\mathrm{X}|\mathrm{Z}} \mid \mathrm{P}_{\mathrm{Z}}\mathrm{P}_{\mathrm{Y}|\mathrm{Z}}) = D(\mathrm{P}_{\mathrm{X}|\mathrm{YZ}}\|\mathrm{P}_{\mathrm{X}|\mathrm{Z}} \mid \mathrm{P}_{\mathrm{YZ}}) \\ &= D(\mathrm{P}_{\mathrm{Y}|\mathrm{XZ}}\|\mathrm{P}_{\mathrm{Y}|\mathrm{Z}} \mid \mathrm{P}_{\mathrm{Z}}\mathrm{P}_{\mathrm{X}|\mathrm{Z}}) = D(\mathrm{P}_{\mathrm{Y}|\mathrm{XZ}}\|\mathrm{P}_{\mathrm{Y}|\mathrm{Z}} \mid \mathrm{P}_{\mathrm{XZ}}). \end{aligned}$$

If X is discrete,

$$\begin{aligned} I(\mathrm{X};\mathrm{Y}) &= D(\mathrm{P}_{\mathrm{X}|\mathrm{Y}}\|\mathrm{P}_{\mathrm{X}} \mid \mathrm{P}_{\mathrm{Y}}) \\ &= \mathbb{E}_{\mathrm{P}_{\mathrm{Y}}}\left[\mathbb{E}_{\mathrm{P}_{\mathrm{X}|\mathrm{Y}}}\Big[\log\frac{p(\mathrm{X}\,|\,\mathrm{Y})}{p(\mathrm{X})}\Big]\right] \\ &= \mathbb{E}\Big[\log\frac{1}{p(\mathrm{X})}\Big] - \mathbb{E}\Big[\log\frac{1}{p(\mathrm{X}\,|\,\mathrm{Y})}\Big] \\ &= H(\mathrm{X}) - H(\mathrm{X}\,|\,\mathrm{Y}). \end{aligned}$$

If $\mathrm{P}_{\mathrm{X}|\mathrm{Y}=y}$ has a density for P_{Y}-almost every y (then so does P_{X}) and $h(\mathrm{X})$, $h(\mathrm{X}\,|\,\mathrm{Y})$ are finite, $I(\mathrm{X};\mathrm{Y}) = h(\mathrm{X}) - h(\mathrm{X}\,|\,\mathrm{Y})$.

If X and Y are both discrete, the entropies and the mutual information satisfy the **Venn diagram** relations



$$H(\mathrm{X},\mathrm{Y}) = H(\mathrm{X}) + H(\mathrm{Y}\,|\,\mathrm{X}) = H(\mathrm{Y}) + H(\mathrm{X}\,|\,\mathrm{Y}) = H(\mathrm{X}\,|\,\mathrm{Y}) + I(\mathrm{X};\mathrm{Y}) + H(\mathrm{Y}\,|\,\mathrm{X}),$$
$$I(\mathrm{X};\mathrm{Y}) = H(\mathrm{X}) - H(\mathrm{X}\,|\,\mathrm{Y}) = H(\mathrm{Y}) - H(\mathrm{Y}\,|\,\mathrm{X}) = H(\mathrm{X}) + H(\mathrm{Y}) - H(\mathrm{X},\mathrm{Y}).$$

The same relations hold with h in place of H if (X,Y) has a pdf and all the terms are finite.

"Chain rule":

$$\begin{aligned} I(\mathrm{X};\mathrm{Y},\mathrm{Z}) &= D(\mathrm{P}_{\mathrm{XYZ}} \| \mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{YZ}}) \\ &= D(\mathrm{P}_{\mathrm{XY}}\mathrm{P}_{\mathrm{Z}|\mathrm{XY}} \| \mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}}\mathrm{P}_{\mathrm{Z}|\mathrm{Y}}) \\ &= D(\mathrm{P}_{\mathrm{XY}} \| \mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}}) + D(\mathrm{P}_{\mathrm{Z}|\mathrm{XY}} \| \mathrm{P}_{\mathrm{Z}|\mathrm{Y}} \mid \mathrm{P}_{\mathrm{XY}}) \\ &= I(\mathrm{X};\mathrm{Y}) + I(\mathrm{X};\mathrm{Z}\,|\,\mathrm{Y}), \end{aligned}$$

and, applying it repeatedly,

$$\begin{aligned} I(\mathrm{X};\mathrm{Y}^n) &= I(\mathrm{X};\mathrm{Y}^{n-1},\mathrm{Y}_n) \\ &= I(\mathrm{X};\mathrm{Y}^{n-1}) + I(\mathrm{X};\mathrm{Y}_n\,|\,\mathrm{Y}^{n-1}) \\ &\;\;\vdots \\ &= \sum_{i=1}^{n} I(\mathrm{X};\mathrm{Y}_i\,|\,\mathrm{Y}^{i-1}). \end{aligned}$$

1.3 Properties

1.3.1 Positivity

- $H(\mathrm{P}) \geq 0$ for discrete P.
- By Jensen's inequality with the convex function $x \mapsto x \log x$,
  $D(\mathrm{P}\|\mathrm{Q}) = \mathbb{E}_{\mathrm{Q}}\Big[\dfrac{p(\mathrm{X})}{q(\mathrm{X})} \log \dfrac{p(\mathrm{X})}{q(\mathrm{X})}\Big] \geq \Big(\mathbb{E}_{\mathrm{Q}}\Big[\dfrac{p(\mathrm{X})}{q(\mathrm{X})}\Big]\Big) \log\Big(\mathbb{E}_{\mathrm{Q}}\Big[\dfrac{p(\mathrm{X})}{q(\mathrm{X})}\Big]\Big) = 0$. Hence
  $$D(\mathrm{P}\|\mathrm{Q}) \geq 0 \qquad \circledast \qquad \text{with equality iff } \mathrm{P} = \mathrm{Q}.$$
  Since $H(\mathrm{P}\|\mathrm{Q}) - H(\mathrm{P}) = D(\mathrm{P}\|\mathrm{Q})$, we get
  $$H(\mathrm{P}) \leq H(\mathrm{P}\|\mathrm{Q}), \quad \forall\, \mathrm{P},\mathrm{Q},$$
  $$H(\mathrm{P}) = \min_{\mathrm{Q}} H(\mathrm{P}\|\mathrm{Q}).$$

- By $\circledast$, $I(\mathrm{X};\mathrm{Y}) \geq 0$, with equality iff $\mathrm{P}_{\mathrm{XY}} = \mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}}$, i.e., X and Y are independent. Similarly, $I(\mathrm{X};\mathrm{Z}\,|\,\mathrm{Y}) \geq 0$, with equality iff $\mathrm{P}_{\mathrm{XZ}|\mathrm{Y}} = \mathrm{P}_{\mathrm{X}|\mathrm{Y}}\mathrm{P}_{\mathrm{Z}|\mathrm{Y}}$ (P_{Y}-a.s.), i.e., X and Z are independent conditionally on Y, i.e., $\mathrm{X} \to \mathrm{Y} \to \mathrm{Z}$ forms a Markov chain.

1.3.2 Conditioning

- $H(\mathrm{X}) \geq H(\mathrm{X}\,|\,\mathrm{Y})$, i.e., $H(\mathrm{P}_{\mathrm{X}}) \geq H(\mathrm{P}_{\mathrm{X}|\mathrm{Y}} \mid \mathrm{P}_{\mathrm{Y}})$: "conditioning reduces entropy", since $H(\mathrm{X}) - H(\mathrm{X}\,|\,\mathrm{Y}) = I(\mathrm{X};\mathrm{Y}) \geq 0$.
- $D(\mathrm{P}_{\mathrm{X}|\mathrm{Y}}\|\mathrm{Q}_{\mathrm{X}|\mathrm{Y}} \mid \mathrm{P}_{\mathrm{Y}}) \geq D(\tilde{\mathrm{P}}_{\mathrm{X}}\|\tilde{\mathrm{Q}}_{\mathrm{X}})$, where $\tilde{\mathrm{P}}_{\mathrm{X}} = \mathbb{E}_{\tilde{\mathrm{Y}}\sim\mathrm{P}_{\mathrm{Y}}}\big[\mathrm{P}_{\mathrm{X}|\mathrm{Y}=\tilde{\mathrm{Y}}}\big] \equiv \mathrm{P}_{\mathrm{X}|\mathrm{Y}} \circ \mathrm{P}_{\mathrm{Y}}$, $\tilde{\mathrm{Q}}_{\mathrm{X}} = \mathbb{E}_{\tilde{\mathrm{Y}}\sim\mathrm{P}_{\mathrm{Y}}}\big[\mathrm{Q}_{\mathrm{X}|\mathrm{Y}=\tilde{\mathrm{Y}}}\big] \equiv \mathrm{Q}_{\mathrm{X}|\mathrm{Y}} \circ \mathrm{P}_{\mathrm{Y}}$.

  Indeed, let $\mathrm{P}_{\mathrm{XY}} = \mathrm{P}_{\mathrm{Y}}\mathrm{P}_{\mathrm{X}|\mathrm{Y}}$ and $\mathrm{Q}_{\mathrm{XY}} = \mathrm{P}_{\mathrm{Y}}\mathrm{Q}_{\mathrm{X}|\mathrm{Y}}$, whose X-marginals are $\tilde{\mathrm{P}}_{\mathrm{X}}$ and $\tilde{\mathrm{Q}}_{\mathrm{X}}$. The chain rule of D, applied in both orders, gives

$$\begin{aligned} D(\mathrm{P}_{\mathrm{X}|\mathrm{Y}}\|\mathrm{Q}_{\mathrm{X}|\mathrm{Y}} \mid \mathrm{P}_{\mathrm{Y}}) &= D(\mathrm{P}_{\mathrm{XY}}\|\mathrm{Q}_{\mathrm{XY}}) = D(\tilde{\mathrm{P}}_{\mathrm{X}}\|\tilde{\mathrm{Q}}_{\mathrm{X}}) + D(\mathrm{P}_{\mathrm{Y}|\mathrm{X}}\|\mathrm{Q}_{\mathrm{Y}|\mathrm{X}} \mid \tilde{\mathrm{P}}_{\mathrm{X}}) \\ &\geq D(\tilde{\mathrm{P}}_{\mathrm{X}}\|\tilde{\mathrm{Q}}_{\mathrm{X}}). \end{aligned}$$

1.3.3 Data processing

For any channel $\mathrm{P}_{\mathrm{Y}|\mathrm{X}} \in \mathcal{P}(\mathcal{Y}|\mathcal{X})$ and any $\mathrm{P}_{\mathrm{X}}, \mathrm{Q}_{\mathrm{X}} \in \mathcal{P}(\mathcal{X})$,

$$D(\mathrm{P}_{\mathrm{X}}\|\mathrm{Q}_{\mathrm{X}}) \geq D(\mathrm{P}_{\mathrm{Y}|\mathrm{X}} \circ \mathrm{P}_{\mathrm{X}}\|\mathrm{P}_{\mathrm{Y}|\mathrm{X}} \circ \mathrm{Q}_{\mathrm{X}}).$$

Indeed, let $\tilde{\mathrm{P}}_{\mathrm{Y}} = \mathrm{P}_{\mathrm{Y}|\mathrm{X}} \circ \mathrm{P}_{\mathrm{X}}$ and $\tilde{\mathrm{Q}}_{\mathrm{Y}} = \mathrm{P}_{\mathrm{Y}|\mathrm{X}} \circ \mathrm{Q}_{\mathrm{X}}$ be the Y-marginals of $\mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}}$ and $\mathrm{Q}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}}$, and $\tilde{\mathrm{P}}_{\mathrm{X}|\mathrm{Y}}$, $\tilde{\mathrm{Q}}_{\mathrm{X}|\mathrm{Y}}$ their conditionals. The chain rule of D, applied in both orders, gives

$$\begin{aligned} D(\mathrm{P}_{\mathrm{X}}\|\mathrm{Q}_{\mathrm{X}}) &= D(\mathrm{P}_{\mathrm{X}}\|\mathrm{Q}_{\mathrm{X}}) + D(\mathrm{P}_{\mathrm{Y}|\mathrm{X}}\|\mathrm{P}_{\mathrm{Y}|\mathrm{X}} \mid \mathrm{P}_{\mathrm{X}}) \\ &= D(\mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}}\|\mathrm{Q}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}|\mathrm{X}}) \\ &= D(\tilde{\mathrm{P}}_{\mathrm{X}|\mathrm{Y}}\tilde{\mathrm{P}}_{\mathrm{Y}}\|\tilde{\mathrm{Q}}_{\mathrm{X}|\mathrm{Y}}\tilde{\mathrm{Q}}_{\mathrm{Y}}) \\ &= D(\tilde{\mathrm{P}}_{\mathrm{Y}}\|\tilde{\mathrm{Q}}_{\mathrm{Y}}) + D(\tilde{\mathrm{P}}_{\mathrm{X}|\mathrm{Y}}\|\tilde{\mathrm{Q}}_{\mathrm{X}|\mathrm{Y}} \mid \tilde{\mathrm{P}}_{\mathrm{Y}}) \\ &\geq D(\tilde{\mathrm{P}}_{\mathrm{Y}}\|\tilde{\mathrm{Q}}_{\mathrm{Y}}). \end{aligned}$$

For mutual information, if $\mathrm{X} \to \mathrm{Y} \to \mathrm{Z}$ is a Markov chain, then $I(\mathrm{X};\mathrm{Z}) \leq I(\mathrm{X};\mathrm{Y})$. Indeed, $I(\mathrm{X};\mathrm{Z}\,|\,\mathrm{Y}) = 0$ for a Markov chain, so

$$\begin{aligned} I(\mathrm{X};\mathrm{Y}) &= I(\mathrm{X};\mathrm{Y}) + \overbrace{I(\mathrm{X};\mathrm{Z}\,|\,\mathrm{Y})}^{0} \\ &= I(\mathrm{X};\mathrm{Y},\mathrm{Z}) \\ &= I(\mathrm{X};\mathrm{Z}) + \underbrace{I(\mathrm{X};\mathrm{Y}\,|\,\mathrm{Z})}_{\geq 0} \\ &\geq I(\mathrm{X};\mathrm{Z}). \end{aligned}$$

1.3.4 Convexity/Concavity

- $\mathrm{P} \mapsto H(\mathrm{P})$ is concave on $\mathcal{P}(\mathcal{X})$. Let $\mathrm{S} \sim \mathrm{P_S} = \mathrm{Bern}(\lambda)$, $\mathrm{P_{X|S=0}} = \mathrm{P}_0$ and $\mathrm{P_{X|S=1}} = \mathrm{P}_1$, so that $\mathrm{P_X} = (1-\lambda)\mathrm{P}_0 + \lambda \mathrm{P}_1$. Then

$$\begin{aligned}\lambda H(\mathrm{P}_1) + (1-\lambda) H(\mathrm{P}_0) &= \sum_{s\in\{0,1\}} p(s)\, H(\mathrm{P_{X|S=}}_s) \\ &= \mathbb{E}_{\bar{\mathrm{S}}\sim\mathrm{Bern}(\lambda)}\big[H(\mathrm{P_{X|S=}}_{\bar{\mathrm{S}}})\big] \\ &= H(\mathrm{X}\,|\,\mathrm{S}) \le H(\mathrm{X}) \equiv H(\mathrm{P_X}) \\ &= H\big((1-\lambda)\mathrm{P}_0 + \lambda\mathrm{P}_1\big),\end{aligned}$$

  since conditioning reduces entropy.

- $\mathrm{P_X} \mapsto I(\mathrm{P_X}, \mathrm{P_{Y|X}})$ is concave on $\mathcal{P}(\mathcal{X})$. With S and X as above, let Y be the output of the channel $\mathrm{P_{Y|X}}$, so that $\mathrm{S} \to \mathrm{X} \to \mathrm{Y}$. Then

$$\begin{aligned}\lambda I(\mathrm{P}_1, \mathrm{P_{Y|X}}) + (1-\lambda) I(\mathrm{P}_0, \mathrm{P_{Y|X}}) &= I(\mathrm{X};\mathrm{Y}\,|\,\mathrm{S}) \le I(\mathrm{X},\mathrm{S};\mathrm{Y}) = I(\mathrm{X};\mathrm{Y}) \\ &= I(\mathrm{P_X}, \mathrm{P_{Y|X}}) = I\big((1-\lambda)\mathrm{P}_0 + \lambda\mathrm{P}_1, \mathrm{P_{Y|X}}\big),\end{aligned}$$

  by the chain rule and since $I(\mathrm{S};\mathrm{Y}\,|\,\mathrm{X}) = 0$.

- $(\mathrm{P},\mathrm{Q}) \mapsto D(\mathrm{P}\|\mathrm{Q})$ is convex on $\mathcal{P}(\mathcal{X}) \times \mathcal{P}(\mathcal{X})$. With S as above, let $\mathrm{P_{X|S=}}_s = \mathrm{P}_s$ and $\mathrm{Q_{X|S=}}_s = \mathrm{Q}_s$, $s \in \{0,1\}$. By "conditioning",

$$\begin{aligned}\lambda D(\mathrm{P}_1\|\mathrm{Q}_1) + (1-\lambda) D(\mathrm{P}_0\|\mathrm{Q}_0) &= D(\mathrm{P_{X|S}}\|\mathrm{Q_{X|S}} \mid \mathrm{P_S}) \\ &\ge D(\tilde{\mathrm{P}}_\mathrm{X}\|\tilde{\mathrm{Q}}_\mathrm{X}) \\ &= D\big((1-\lambda)\mathrm{P}_0 + \lambda\mathrm{P}_1 \,\big\|\, (1-\lambda)\mathrm{Q}_0 + \lambda\mathrm{Q}_1\big).\end{aligned}$$

- $\mathrm{P_{Y|X}} \mapsto I(\mathrm{P_X}, \mathrm{P_{Y|X}})$ is convex on $\mathcal{P}(\mathcal{Y}|\mathcal{X})$. This follows from the joint convexity of D: for a fixed $\mathrm{P_X}$, both arguments of D in $I(\mathrm{P_X}, \mathrm{P_{Y|X}}) = D(\mathrm{P_X P_{Y|X}} \| \mathrm{P_X P_Y})$, where $\mathrm{P_Y} = \mathrm{P_{Y|X}} \circ \mathrm{P_X}$, are linear in $\mathrm{P_{Y|X}}$. Explicitly, let $\mathrm{P_{Y|X}} = (1-\lambda)\mathrm{P}^{(0)}_{\mathrm{Y|X}} + \lambda \mathrm{P}^{(1)}_{\mathrm{Y|X}}$ with $\mathrm{P}^{(0)}_{\mathrm{Y|X}}, \mathrm{P}^{(1)}_{\mathrm{Y|X}} \in \mathcal{P}(\mathcal{Y}|\mathcal{X})$. Then $\mathrm{P_Y} = (1-\lambda)\mathrm{P}^{(0)}_\mathrm{Y} + \lambda \mathrm{P}^{(1)}_\mathrm{Y}$, where $\mathrm{P}^{(s)}_\mathrm{Y} := \mathrm{P}^{(s)}_{\mathrm{Y|X}} \circ \mathrm{P_X}$, and

$$\begin{aligned}&\lambda I(\mathrm{P_X}, \mathrm{P}^{(1)}_{\mathrm{Y|X}}) + (1-\lambda) I(\mathrm{P_X}, \mathrm{P}^{(0)}_{\mathrm{Y|X}}) \\ &\quad= \lambda D(\mathrm{P_X P}^{(1)}_{\mathrm{Y|X}} \| \mathrm{P_X P}^{(1)}_\mathrm{Y}) + (1-\lambda) D(\mathrm{P_X P}^{(0)}_{\mathrm{Y|X}} \| \mathrm{P_X P}^{(0)}_\mathrm{Y}) \\ &\quad\ge D\big((1-\lambda)\mathrm{P_X P}^{(0)}_{\mathrm{Y|X}} + \lambda \mathrm{P_X P}^{(1)}_{\mathrm{Y|X}} \,\big\|\, (1-\lambda)\mathrm{P_X P}^{(0)}_\mathrm{Y} + \lambda \mathrm{P_X P}^{(1)}_\mathrm{Y}\big) \\ &\quad= D(\mathrm{P_X P_{Y|X}} \| \mathrm{P_X P_Y}) \\ &\quad= I(\mathrm{P_X}, \mathrm{P_{Y|X}}).\end{aligned}$$

1.4 Maximum entropy

The method relies on the following equivalence: for any P and Q,

$$D(\mathrm{P}\|\mathrm{Q}) \ge 0 \quad \iff \quad H(\mathrm{P}) \le H(\mathrm{P}\|\mathrm{Q}).$$

1.4.1 Finite alphabet

Let $|\mathcal{X}| = M$. Take $\mathrm{Q} \in \mathcal{P}(\mathcal{X})$ uniform, $q(x) = \dfrac{1}{M}$. Then

$$\begin{aligned} H(\mathrm{P}) \le H(\mathrm{P}\|\mathrm{Q}) &= \mathbb{E}_{\mathrm{P}}[\log M] \\ &= \log M, \end{aligned}$$

with equality iff $\mathrm{P} = \mathrm{Q}$ is uniform.

1.4.2 Continuous alphabet

Which P with $\mathbb{E}_{\mathrm{P}}[\mathrm{X}^2] \le \sigma^2$ maximizes $h(\mathrm{P})$? Take $\mathrm{Q} = \mathcal{N}(0, \sigma^2)$. Then

$$\begin{aligned} h(\mathrm{P}) \le h(\mathrm{P}\|\mathrm{Q}) &= \mathbb{E}_{\mathrm{P}}\Big[\log\sqrt{2\pi\sigma^2} + \frac{\mathrm{X}^2}{2\sigma^2}\log e\Big] \\ &\le \log\sqrt{2\pi\sigma^2} + \frac{1}{2}\log e = \log\sqrt{2\pi e\sigma^2} \\ &= h(\mathrm{Q}), \end{aligned}$$

with equality iff $\mathrm{P} = \mathrm{Q} = \mathcal{N}(0, \sigma^2)$.

1.4.3 General case

Which P with $\mathbb{E}_{\mathrm{P}}\big[c(\mathrm{X})\big] \le a$ maximizes $h(\mathrm{P})$, for a cost function $c \ge 0$? Take

$$q(x) = \frac{e^{-\beta c(x)}}{\int_{\mathcal{X}} e^{-\beta c(x)}\,\mathrm{d}x} = \frac{1}{Z}\,e^{-\beta c(x)},$$

with $\beta \ge 0$ chosen such that $\mathbb{E}_{\mathrm{Q}}\big[c(\mathrm{X})\big] = a$ (or $\beta = 0$ if $\mathbb{E}_{\mathrm{Q}}\big[c(\mathrm{X})\big] \le a$ already holds for $\beta = 0$). Then

$$\begin{aligned} h(\mathrm{P}) \le h(\mathrm{P}\|\mathrm{Q}) &= \mathbb{E}_{\mathrm{P}}\big[\log Z + \beta\, c(\mathrm{X})\log e\big] \\ &\le \mathbb{E}_{\mathrm{Q}}\big[\log Z + \beta\, c(\mathrm{X})\log e\big] \\ &= h(\mathrm{Q}), \end{aligned}$$

since $\beta\,\mathbb{E}_{\mathrm{P}}\big[c(\mathrm{X})\big] \le \beta a = \beta\,\mathbb{E}_{\mathrm{Q}}\big[c(\mathrm{X})\big]$, with equality iff $\mathrm{P} = \mathrm{Q}$. This Q is called the **Gibbs distribution**, β the **inverse temperature**, and Z the **partition function**. The same argument holds for a discrete $\mathcal{X}$, with a sum in place of the integral and H in place of h. Both examples above are Gibbs distributions: $c(x) = 0$ (no constraint) with $\beta = 0$ on a finite $\mathcal{X}$ gives the uniform Q, and $c(x) = x^2$ with $\beta = \frac{1}{2\sigma^2}$ gives $\mathrm{Q} = \mathcal{N}(0, \sigma^2)$ (on $\mathcal{X} = \mathbb{R}$, $Z < \infty$ forces $\beta > 0$).

Summary

- Every distribution is defined explicitly: an input distribution $\mathrm{P_X} \in \mathcal{P}(\mathcal{X})$, a channel $\mathrm{P_{Y|X}} \in \mathcal{P}(\mathcal{Y}|\mathcal{X})$, the joint distribution $\mathrm{P_X P_{Y|X}}$ and the output distribution $\mathrm{P_{Y|X}} \circ \mathrm{P_X}$, a mixture.

- Jensen's inequality: $\mathbb{E}\big[f(\mathrm{X})\big] \geq f\big(\mathbb{E}[\mathrm{X}]\big)$ for a convex f.
- Information measures: entropy $H(\mathrm{P})$, cross-entropy $H(\mathrm{P}\|\mathrm{Q})$, divergence $D(\mathrm{P}\|\mathrm{Q}) = H(\mathrm{P}\|\mathrm{Q}) - H(\mathrm{P})$ and mutual information $I(\mathrm{X};\mathrm{Y}) = D(\mathrm{P}_{\mathrm{XY}}\|\mathrm{P}_{\mathrm{X}}\mathrm{P}_{\mathrm{Y}})$. Their conditional versions are averages over the conditioning variable, and each measure has a chain rule.
- $D(\mathrm{P}\|\mathrm{Q}) \geq 0$, with equality iff $\mathrm{P} = \mathrm{Q}$. Hence $I(\mathrm{X};\mathrm{Y}) \geq 0$, $H(\mathrm{P}) \leq H(\mathrm{P}\|\mathrm{Q})$, conditioning reduces entropy and increases divergence, and processing reduces divergence and mutual information (data processing).
- $H(\mathrm{P})$ is concave in P; $I(\mathrm{P}_{\mathrm{X}}, \mathrm{P}_{\mathrm{Y}|\mathrm{X}})$ is concave in P_{X} and convex in $\mathrm{P}_{\mathrm{Y}|\mathrm{X}}$; $D(\mathrm{P}\|\mathrm{Q})$ is jointly convex.
- Maximum entropy under $\mathbb{E}_{\mathrm{P}}\big[c(\mathrm{X})\big] \leq a$: the Gibbs distribution $q(x) = e^{-\beta c(x)}/Z$, e.g., the uniform distribution on a finite alphabet, and $\mathcal{N}(0, \sigma^2)$ under $\mathbb{E}_{\mathrm{P}}[\mathrm{X}^2] \leq \sigma^2$.